

Two-Stage Cascaded Information Retrieval and LLM Orchestration for Automated Resume Screening

Capstone Project CS[02]: Technical Report

1 Introduction and Problem Statement

Manual resume screening across large applicant pools is slow, subjective, and operationally inefficient. Traditional lexical matching tools fail against keyword stuffing, where applicants artificially match job descriptions without having verified experience. Conversely, evaluating entire pools with commercial Large Language Models (LLMs) incurs high latency, substantial API costs, and vulnerability to indirect prompt injections.

To resolve these challenges, this project presents an automated, two-stage Information Retrieval (IR) screening pipeline orchestrated via LangChain [1]. The system provides two operational modes:

- **Mode A (Local Neural Engine):** Runs completely offline on commodity CPU hardware using open-source transformers, incurring zero operational API costs.
- **Mode B (Generative LLM Engine):** Employs an external LLM via LangChain with Pydantic structured output validation to synthesize explainable recruiter audit scorecards.

2 System Architecture and Model Justifications

The pipeline is constructed as an acyclic graph via LangChain Expression Language (LCEL) with strict schema validation contracts across all intermediate representations.

2.1 Document Ingestion and Security Guardrails

Resumes (PDF/TXT) are parsed and normalized via Unicode NFKC decomposition, stripping non-printable control characters ($\mathcal{C}_c, \mathcal{C}_f$). A compiled, case-insensitive regex engine scans incoming text for adversarial injection vectors (e.g., system command overrides, delimiters, or score overrides). Detected payloads are redacted into benign tokens, raising an alert flag that penalizes the composite score by 50%.

2.2 Stage 1: Coarse Hybrid Retrieval Model

Filters large applicant pools down to a high-recall shortlist in sub-linear vector lookup time.

- **Selected Model:** BAAI/bge-base-en-v1.5 (Bi-Encoder, 768 dimensions, 512 tokens) [2].
- **Technical Justification:** Job descriptions and resumes are structurally asymmetric in syntax and length. While symmetric models suffer under length disparity, BGE prepends an asymmetric search instruction prefix to the query vector space ($\mathbf{q} \in \mathbb{R}^{768}$) while encoding raw resumes ($\mathbf{d} \in \mathbb{R}^{768}$), achieving top-tier retrieval on the MTEB benchmark.

2.3 Stage 2: Deep Cross-Encoder and Generative Reranking

Refines Stage 1 outputs using joint cross-attention or generative semantic reasoning:

- **Mode A Neural Model:** BAAI/bge-reranker-base [2]. Jointly encodes the concatenated pair $[\mathbf{q}; \mathbf{d}]$ with all-to-all cross-attention. Trained on hard negatives, it penalizes superficial keyword matching that deceives bi-encoder cosine similarity.
- **Mode B Generative Model:** openai/gpt-oss-20b (via Groq LPU) with gpt-4o-mini fallback.
 - *Zero Marginal Cost:* Accessible over Groq’s developer free tier, eliminating testing costs.

- *LPU Hardware Acceleration*: Operating as a 21B Mixture-of-Experts architecture ($\approx 3.6B$ active parameters/token) served on Groq LPUs, it completes inference in under 450 ms per resume.
- *Deterministic Schema Conformance*: Native integration with LangChain’s `.with_structured_output()` ensures valid Pydantic JSON decoding with zero hallucinated keys.
- **OpenAI Fallback**: An adaptive client factory routes to `gpt-4o-mini` if standard OpenAI keys (`sk-...`) are detected instead of Groq keys (`gsk-...`), ensuring vendor independence.

3 Mathematical Formulation of Candidate Scoring

3.1 Stage 1: Convex Hybrid Combination

For query q and candidate document d_i , dense semantic similarity ($\sigma_{\text{dense}} = \frac{q \cdot d_i}{\|q\| \|d_i\|} \times 100$) and exact mandatory skill overlap ($\sigma_{\text{lex}} = \frac{|S_{\text{cand}} \cap S_{\text{jd}}|}{|S_{\text{jd}}|} \times 100$) are blended into a coarse retrieval score:

$$S_{\text{Stage1}} = 0.60 \cdot \sigma_{\text{dense}} + 0.40 \cdot \sigma_{\text{lex}} \quad (1)$$

3.2 Stage 2: Multi-Attribute Rubric and Non-Linear Penalization

Stage 2 evaluates candidate profiles across four calibrated dimensions:

$$R_{\text{Stage2}} = 0.40 \cdot S_{\text{tech}} + 0.35 \cdot S_{\text{exp}} + 0.15 \cdot S_{\text{domain}} + 0.10 \cdot S_{\text{edu}} \quad (2)$$

where S_{domain} and S_{edu} denote domain and academic fit. To counter keyword stuffers who artificially inflate S_{tech} , S_{exp} enforces a **quadratic tenure penalty** when verified experience (T_{cand}) is below the required experience (T_{req}):

$$S_{\text{exp}} = \begin{cases} \left(\frac{T_{\text{cand}}}{T_{\text{req}}}\right)^2 \times 100, & \text{if } T_{\text{cand}} < T_{\text{req}} \\ \min\left(100.0, 75.0 + 50.0 \left(\frac{T_{\text{cand}}}{T_{\text{req}}} - 1.0\right)\right), & \text{if } T_{\text{cand}} \geq T_{\text{req}} \end{cases} \quad (3)$$

Under linear scaling, an applicant with 1.0 year of tenure for a 4.0-year role receives 25.0%; the quadratic function drops this to $(0.25)^2 \times 100 = 6.25\%$. Furthermore, an **Anti-Keyword-Stuffing Gate** suppresses lexical inflation:

$$R_{\text{calibrated}} = \begin{cases} R_{\text{Stage2}} \times 0.40, & \text{if } \frac{|S_{\text{cand}} \cap S_{\text{jd}}|}{|S_{\text{jd}}|} \geq 0.70 \text{ and } T_{\text{cand}} < 0.60 T_{\text{req}} \\ R_{\text{Stage2}}, & \text{otherwise} \end{cases} \quad (4)$$

3.3 Final Composite Score and Security Gating

The global ranking combines both stages, scaled by threat detection indicator $\mathbb{I}_{\text{threat}} \in \{0, 1\}$:

$$S_{\text{final}} = (0.20 \cdot \sigma_{\text{dense}} + 0.10 \cdot \sigma_{\text{lex}} + 0.70 \cdot R_{\text{calibrated}}) \times (1.0 - 0.50 \cdot \mathbb{I}_{\text{threat}}) \quad (5)$$

4 Experimental Testing, Results, and Verification

The framework was evaluated across a standardized benchmark corpus of **1,000 resumes** spanning **10 distinct job domain suites** [4]. Each suite contained 100 resumes divided into four cohorts: qualified seniors ($n = 20$), borderline mid-levels ($n = 25$), junior keyword stuffers ($n = 30$), and unrelated/adversarial profiles ($n = 25$).

Table 1: Empirical Benchmark Metrics Across Both Operational Modes

Benchmark Suite (100 CVs Each)	P@20 (A / B)	MRR (A / B)	Avg Stuffer (A / B)	Threats
Suite 01: Senior Backend Engineer	100.0% / 100.0%	1.00 / 1.00	#75.5 / #38.5	10 / 10
Suite 02: ML Research Engineer	85.0% / 100.0%	1.00 / 1.00	#75.5 / #38.0	10 / 10
Suite 03: DevOps & Platform	95.0% / 100.0%	1.00 / 1.00	#75.5 / #37.5	10 / 10
Suite 04: Data Platform Engineer	85.0% / 100.0%	1.00 / 1.00	#75.5 / #39.0	10 / 10
Suite 05: Lead Frontend Engineer	65.0% / 95.0%	1.00 / 1.00	#75.5 / #38.5	10 / 10
Suite 06: Cybersecurity Analyst	40.0% / 90.0%	0.17 / 1.00	#43.9 / #40.0	10 / 10
Suite 07: Full Stack Engineer	95.0% / 100.0%	1.00 / 1.00	#75.5 / #37.0	10 / 10
Suite 08: Embedded Systems	80.0% / 100.0%	1.00 / 1.00	#75.5 / #38.5	10 / 10
Suite 09: Database Administrator	85.0% / 100.0%	1.00 / 1.00	#75.5 / #38.0	10 / 10
Suite 10: NLP & Agentic AI Architect	100.0% / 100.0%	1.00 / 1.00	#75.5 / #38.0	10 / 10
Aggregate Benchmark (1,000 CVs)	83.00% / 98.50%	0.9167 / 1.0000	#72.3 / #38.3	100 / 100

4.1 Comparative Analysis of Experimental Findings

- **Precision@20 & Ranking Quality:** Mode A maintained an 83.00% Mean P@20 and 0.9167 MRR across 1,000 CVs on local CPU. Mode B achieved 98.50% Mean P@20 and 1.0000 MRR, isolating verified seniors at ranks #1–#20 (scores 72.44–84.27).
- **Keyword Stuffer Demotion:** Quadratic tenure scaling demoted stuffers to average rank #72.3 in Mode A. Mode B autoregressively identified superficial listings, reducing Stage 2 scores from 85.0% to 19.5%–40.5% (ranks #21–#46).
- **Adversarial Neutralization:** Guardrails intercepted 100% (100/100) of prompt injection payloads, enforcing 50% score penalties and relegating attackers to ranks #91–#100 (scores 7.54–12.74).
- **Latency Profile:** Mode A completed evaluations in 65 ms/resume on CPU within a 1.1 GB memory footprint. Mode B produced structured recruiter justifications in 420 ms/resume on Groq LPUs.

5 Conclusion

This project demonstrates an auditable, two-stage IR resume screening pipeline orchestrated via LangChain. By combining asymmetric bi-encoders, hard-negative cross-encoders, and schema-constrained generative reasoning, the system eliminates keyword inflation while neutralizing prompt injections. The dual-mode design balances operational flexibility: Mode A provides cost-free, high-throughput offline screening on commodity CPUs, while Mode B offers rich, explainable recruiter scorecard audits.

References

- [1] H. Chase, “LangChain: Building applications with LLMs through composability,” 2023. [Online]. Available: <https://github.com/langchain-ai/langchain>
- [2] Beijing Academy of Artificial Intelligence (BAAI), “BAAI General Embedding (BGE) Models: Bi-Encoder and Reranker Series,” 2023. [Online]. Available: <https://huggingface.co/BAAI>
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- [4] Project Repository & 1,000-CV Empirical Evaluation Suite, 2026. [Online]. Available: <https://github.com/SHIVANGIJOSHI/cv-sorting-pipeline-cs02>